RSC Advances

PAPER

Au–Ag bimetallic nanodendrite synthesized *via* simultaneous co-electrodeposition and its application as a SERS substrate

Cite this: *RSC Adv.*, 2014, **4**, 3929

Tran Ngoc Huan,[a] Saetbyeol Kim,[a] Pham Van Tuong[b] and Hoeil Chung*[a]

Received 6th September 2013
Accepted 20th November 2013

DOI: 10.1039/c3ra44916e

www.rsc.org/advances

We presented a novel Au–Ag bimetallic nanodendrite constructed through simple one-step co-electrodeposition on a Pt surface in the presence of iodide ions as a co-reagent. Simultaneously, we maneuvered to generate hydrogen bubbles from the same surface by the reduction of proton to drive the growth of deposition into the nanodendrite network structure. Au–Ag bimetallic nanodendrites composed of different atomic percentages of Au and Ag (0.00–5.12% Ag) were electrochemically synthesized by varying mole percentages of Au^{3+} and Ag^{+} in solutions. When atomic Ag concentration was from 0.39 to 3.84%, characteristic dendrite structure with high crystallinity was clearly observed; while, an imperfect nanodendrite structure became present when the atomic Ag concentration was over 3.84%. Further, the bimetallic nanodendrites with different Ag concentrations were used as a substrate for surface-enhanced Raman scattering (SERS) measurement. The SERS intensity of 2-naphthalenethiol was maximized when the nanodendrite containing 3.84% Ag was used. Incorporation of better plasmonic metal, Ag, into the nanodendrite structure was obviously advantages in enhancing SERS efficiency; while, maintaining a good bimetallic nanodendrite structure was a prerequisite for optimal SERS measurement.

Introduction

Metal composite nanostructures have received great attention because these provide enhanced local electric field, high catalytic activity, and high conductivity, so making them versatile for use in surface-enhanced Raman scattering (SERS) based measurements and as components of electro-catalysts, fuel cells, and electrochemical sensors.[1–4] In particular, Au–Ag bimetallic nanostructures with different shapes have been demonstrated to enhance SERS signal by combining individual advantages of Ag and Au as a SERS substrate. In most cases, a galvanic displacement reaction has been frequently adopted to synthesize Au–Ag nanostructures.[3–7]

Netzer *et al.* synthesized Au–Ag bimetallic porous nanowires by a galvanic displacement reaction using Ag nano-rods as a starting material (or pre-prepared substrate).[3] Similarly, octahedral Au–Ag nanoframes were constructed from Ag nanoparticles *via* a modified galvanic reaction.[5] In addition, several Au–Ag bimetallic structures fabricated by various methods have been demonstrated; for example, Au–Ag segmented nanorods in alumina film prepared by a one-step galvanic displacement reaction,[4] a nanostructure of assembled Au nanoparticles on Ag spheres,[7] Au on Ag nanorods,[8] nanoporous Au films containing various amounts of residual Ag controlled by de-alloying/annealing,[9] and homogeneous Au–Ag nano-alloys of various compositions synthesized with the aid of femto-second laser excitation.[10]

Among the previously reported Au–Ag bimetallic nanostructures, there are no prior reports on Au–Ag bimetallic nanodendrite structures. Dendrites have a hierarchical structure with high population of edges as well as acute angles in terraces; therefore, there are many hot spots within the structure, which could enhance the sensitivity of SERS measurement.[11–14] In addition, a dendritic structure has a large surface area, advantageous to improve the sensitivity of electrochemical measurement.[15–17] Therefore, a bimetallic nanodendrite structure could be a promising SERS substrate as well as platform in electrochemical sensing.

The only bimetallic nanodendrite structure reported so far is Ag–Cu nanodendrite structure constructed by a galvanic displacement reaction between $AgNO_3$ and Cu substrate.[12] In comparison with the Ag nanodendrite substrate, the Ag–Cu bimetallic nanodendrites showed enhanced SERS properties. Rather than Ag–Cu nanodendrite, Au–Ag bimetallic nanodendrites could be more advantageous for SERS applications, since Au is a more effective plasmonic material than Cu and chemical conjugation of sensing/detection elements on Au surface is facile and well developed. In addition, most of the

[a]*Department of Chemistry, College of Natural Sciences, Hanyang University, Seoul, 133-791, Korea. E-mail: hoeil@hanyang.ac.kr; Fax: +82-2-2299-0762; Tel: +82-2-2220-0937*

[b]*Department of Physics, College of Natural Sciences, Hanyang University, Seoul, 133-791, Korea*

This journal is © The Royal Society of Chemistry 2014

bimetallic nanostructures introduced above have been constructed using a two-step preparation process involving initial preparation of a metal substrate and then incorporation of a second metal on the substrate *via* a galvanic displacement reaction. From a practical viewpoint, a one-step method for synthesis of Au–Ag bimetallic nanostructures is simple and advantageous.

In this publication, we report an effective electrochemical method for synthesis of Au–Ag bimetallic nanodendrites through one-step co-electrodeposition. The method is based on the simultaneous electrodeposition of both Au and Ag on a Pt surface under the presence of iodide ions as a co-reagent. Hydrogen bubbles generating from the same surface drive the growth of bimetallic nanodendrite network structures as described in our previous publication.[16] Au–Ag bimetallic nanodendrites composed of different atomic percentages of Au and Ag (0.00–5.12% Ag) were constructed by the co-deposition and then their structures were characterized using relevant analytical tools. Finally, SERS spectra of 2-naphthalenethiol (2-NpSH) were obtained using these nanodendrites and the variation of resulting intensities was examined in relationship with Ag concentration as well as structural perfectness of the nanodendrite.

Experimental section

Chemicals and electrochemical instrumentations

All chemicals including $HAuCl_4$, (99.999%), KI, NH_4Cl, $AgNO_3$ (99.9%), and 2-NpSH were purchased from Sigma-Aldrich. The electrochemical synthesis of Au–Ag bimetallic nanodendrite structure was performed on a galvanostat (Epsilon, Bioanalytical Chemistry Systems, Inc, USA). Initially, to synthesize Au nanodendrite network structure, a Pt electrode was immersed in a solution composed of 13.33 mM $HAuCl_4$, 0.5 M H_2SO_4 and 2.5 M NH_4Cl. The concentration of KI was maintained at 1.5 mM as described in a previous study.[16] Then, a current of 50 mA was applied for 160 seconds for electrodeposition using a galvanostat. To construct Au–Ag bimetallic nanodendrites with different atomic percentages, the molar percentage of Ag^+ in the solution was varied from 1.0 to 8.0% (increments of 1.0%) relative to that of Au^{3+}, while the sum of Au^{3+} and Ag^+ concentrations was maintained at 13.33 mM. For co-electrodeposition of both metals, a current of 50 mA was also applied to these solutions for 160 seconds at room temperature.

Characterization of nanodendrites and SERS measurement

All electrochemical measurements were performed at room temperature using a standard cell with a three electrode arrangement in which Ag/AgCl and Pt wire were used as reference and counter electrodes, respectively. The potential measurement was performed relative to the Ag/AgCl reference electrode, and all data were recorded using a potentiostat (eDAQ, Model PowerLab/4SP, Denistone East, NSW 2112, Australia). The collection of cyclic voltammograms (CV) of the synthesized nanodendrites was performed relative to the Ag/AgCl reference electrode and all data were recorded using the potentiostat.

Scanning electron microscope (SEM) images and energy-dispersive X-ray (EDX) spectra were acquired using a Hitachi S-4800 scanning electron microscope. Transmission electron microscope (TEM) and high-resolution TEM (HRTEM) images were obtained on a JEM-2100F transmission electron microscope (JEOL, Japan) with an accelerating voltage of 200 kV. X-ray diffraction (XRD) was measured using a D/MAX Rint 2000 diffractometer (Rigaku, Japan) with CuK_α radiation (λ = 1.54178 Å, 40 kV and 200 mA). X-ray photoelectron spectroscopy (XPS) spectra (ESCALAB 220, VG Microtech, UK) of the samples were collected with a Mg K_α X-ray source. SERS spectra were collected using a dispersive Raman microscope equipped with a diode laser (λ_{ex} = 785 nm), a thermoelectrically cooled (−40 °C) charge-coupled detection (CCD) detector, and a holographic grating (Kaiser Optical Inc., USA). 2-NpSH-bound nanodendrites were positioned on a microscope stage and the laser beam was focused using an objective lens to collect SERS spectra with a spectral resolution of 4 cm^{-1}.

Results and discussions

Electrochemical synthesis of Au–Ag bimetallic nanodendrite and characterization

In the synthesis of Ag-containing bimetallic structures, anions such as Cl^- and SO_4^{2-} in a solution could form non-soluble precipitating salts with Ag^+, thereby resulting in imperfect nanostructures. To resolve this problem, $AgNO_3$ was initially diluted in NH_4Cl–NH_4OH buffer before the electrodeposition to drive the formation of the $Ag(NH_3)_2^+$ complex, so preventing Ag^+ ions from forming complexes with other anions. In addition, the concentration of NH_4^+ was kept high (2.5 M) to generate more NH_3 to enhance the formation of $Ag(NH_3)_2^+$.

During the synthesis, three different reactions could possibly occur simultaneously: electrodeposition of both Au and Ag, and galvanic reaction between Ag and $AuCl_4^-$ ($3Ag + AuCl_4^- \rightarrow Au + 3AgCl + Cl^-$). Because E^0 of the $AuCl_4^-$/Au pair (0.99 V *vs.* SHE) is higher than E^0 of the Ag^+/Ag pair (0.80 V *vs.* SHE), the galvanic reaction occurred spontaneously. However, the rate of this reaction at room temperature was much slower than that of electrochemical deposition when high current (such as 50 mA in this study) was applied, therefore it did not seriously interfere with the co-electrodeposition of Au and Ag.[7]

After the synthesis, the atomic percentages of Au and Ag in the bimetallic nanodendrites were determined using EDX analysis by calculating ratios of the corresponding peaks. The atomic percentages of Ag in the nanodendrites prepared using solutions containing 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, and 8.0% Ag^+ (percent in molar concentration relative to Au^{3+}) were 0.16, 0.39, 1.10, 1.82, 2.83, 3.84, 4.48, and 5.12%, respectively. As expected, the relative percentage of Ag in the structures was lower than that of Ag^+ in the corresponding solution used for the electro-deposition, since the deposition rates of Au and Ag were different. From this point on, the atomic percentages of Au and Ag in the nanodendrites are designated in the form

 This journal is © The Royal Society of Chemistry 2014

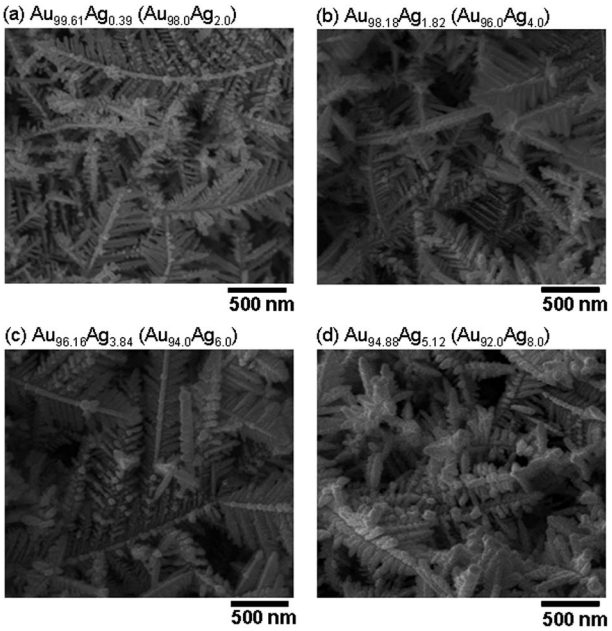

Fig. 1 SEM images of Au–Ag bimetallic nanodendrites with four different atomic Au–Ag percentages. The notation in the parentheses indicates the molar percentage of Au^{3+} and Ag^{+} in the solutions used for co-electrodeposition.

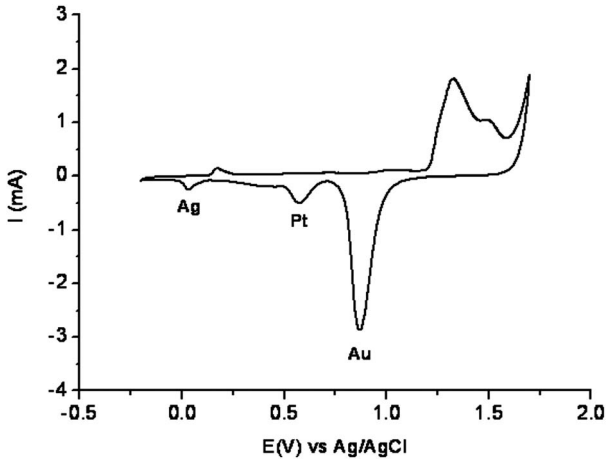

Fig. 2 Cyclic voltammogram (CV) acquired for the $Au_{98.18}$–$Ag_{1.82}$ bimetallic nanodendrite (constructed on a Pt electrode) in a solution of 0.5 M H_2SO_4.

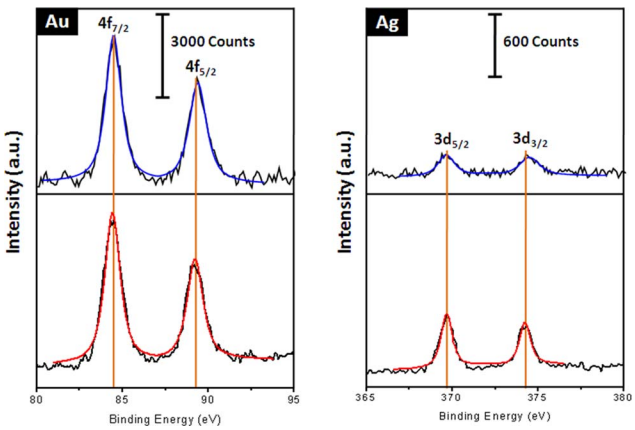

Fig. 3 XPS spectra of Au–Ag bimetallic synthesized from 2 different solutions containing different molar percentage of Au^{3+} and Ag^{+}; 96.0% Au^{3+} : 4.0% Ag^{+} (top) and 92.0% Au^{3+} : 8.0% Ag^{+} (bottom).

$Au_{98.18}$–$Ag_{1.82}$, which indicates the nanodendrite composed of 98.18% Au and 1.82% Ag.

Fig. 1 shows SEM images of four Au–Ag bimetallic nanodendrites prepared by varying the concentrations of Au^{3+} and Ag^{+} for electrodeposition. The concentration notation in parentheses indicates the percentage in molar concentrations of Au^{3+} and Ag^{+} in the corresponding solution for the electrodeposition. Dendrite structures were clearly observed for the three samples containing 0.39–3.84% Ag (Fig. 1(a)–(c)). In general, the growth of Ag itself by electrodeposition tends to result in undefined nanostructures as reported;[18] therefore, by increasing the Ag^{+} concentration, imperfect Au–Ag bimetallic nanodendrite structures would be produced. As found in Fig. 1(d), the bimetallic nanodendrite containing 4.48% Ag shows a less characteristic dendritic appearance with blunt edges. When the molar percentage of Ag^{+} in a solution was further increased, such as up to 14%, the dendrite structure disappeared completely (the corresponding image is not shown).

To simply confirm the co-deposition of Au and Ag, cyclic voltammogram (CV) was also acquired for the $Au_{98.18}$–$Ag_{1.82}$ bimetallic nanodendrite (constructed on a Pt electrode) in 0.5 M H_2SO_4 as shown in Fig. 2. Reduction peaks of Au and Ag were clearly observed, confirming a bimetallic composition. The peak intensity of Ag was substantially lower than that of Au due to the low atomic percentage of Ag (1.82%) in the nanodendrite. The result clearly demonstrates the bimetallic composition of Au and Ag.

Along with the identification of bi-metal deposition by CV, atomic composition in the dendrites was also analyzed using XPS. Fig. 3 shows XPS spectra of two Au–Ag bimetallic nanodendrites prepared from two different electrodeposition solutions; 96.0% Au^{3+} : 4.0% Ag^{+} (top) and 92.0% Au^{3+} : 8.0% Ag^{+} (bottom). Left and right spectra show the peaks of Au and Ag, respectively. For Au, two peaks corresponding $4f_{7/2}$ and $4f_{5/2}$ are apparent at 84.4 and 88.3 eV, respectively. Two Ag peaks at 370.1 and 374.6 eV corresponding to $3d_{5/2}$ and $3d_{3/2}$, respectively, are distinct; while, their intensity is lower due to the lower concentration. With the increase of Ag^{+} concentration for the electrodeposition, the intensities of Ag and Au peaks increase and decreases, respectively. The atomic Ag percentages of two bimetallic nanodendrites calculated using the corresponding XPS spectra by Gaussian fitting were 1.90 and 5.21%, agreeable with the results obtained using EDX (1.82 and 5.12%, respectively).

For more detail characterization of the bimetallic nanodendrites, the corresponding TEM images and XRD spectra were examined. A TEM image highlighting a single $Au_{98.18}$–$Ag_{1.82}$ nanodendrite is shown in Fig. 4(a). Terraces are clearly

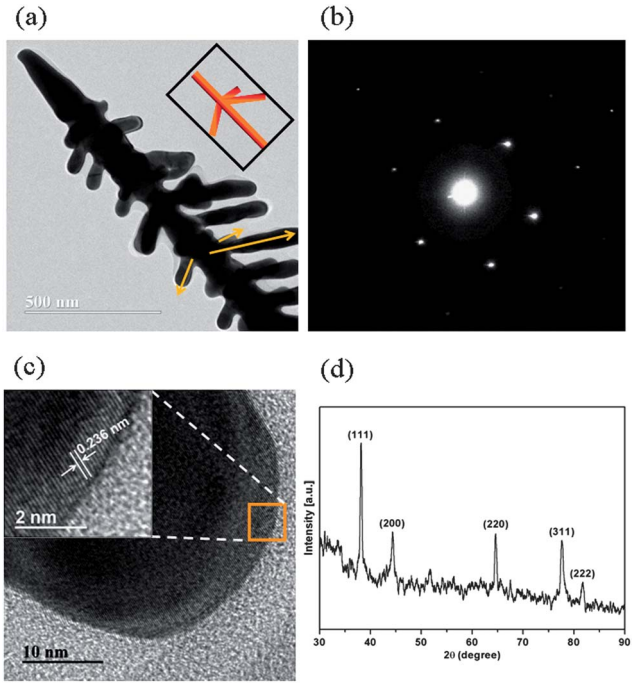

Fig. 4 TEM images of a single $Au_{98.18}$–$Ag_{1.82}$ nanodendrite (a). The orientation of the tips is shown graphically in the inset. Selected area electron diffraction (SAED) pattern of the nanodendrite (b) and HRTEM image of the end of a tip (c) are shown. A magnified image of the far end of the tip (designated by the square) is presented (c). XRD spectrum of the corresponding bimetallic nanodendrite revealing a crystalline structure is also shown (d).

observed along the dendrite column and the tips are three-dimensionally oriented, as indicated by the arrows. The orientation of the tips was graphically regenerated as shown in the inset. The selected area electron diffraction (SAED) pattern of this nanodendrite is shown in Fig. 4(b). A circular diffraction pattern is clear, indicating a highly crystalline structure. HRTEM image on the end of a tip of the nanodendrite is shown in Fig. 4(c). The magnified image of the far end (designated by the square) is also shown. The dark and bright images correspond to the tip and the bare TEM substrate, respectively. In the magnified image, the parallel lines indicate uni-directional growth of the Au–Ag crystalline structure in the tip with an inter-lattice distance of 0.236 nm. Five distinct peaks are observable at 38.23° (111), 44.42° (200), 64.73° (220), 77.71° (311), and 82.41° (222) in the XRD spectrum (Fig. 4(d)). This result again indicates a crystalline structure with a face-centered cubic (fcc) and lattice (unit cell), consistent with the SAED pattern.

Simulation of electromagnetic field enhancement in a nanodendrite structure

For further usage of the nanodendrites especially in SERS applications, it is beneficial to investigate the induced electromagnetic (EM) field in the bimetallic nanodendrite under excitation of laser. Recently, a numerical simulation using the finite-difference time-domain (FDTD) method has been useful

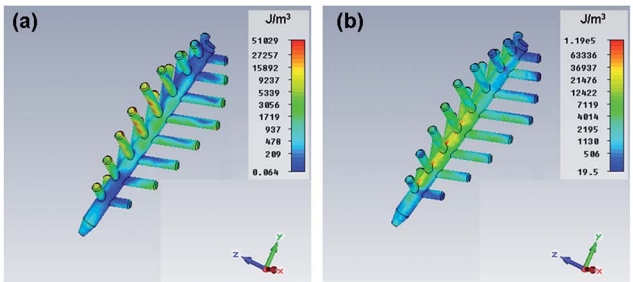

Fig. 5 Distribution of induced near-field electrical (a) and magnetic (b) field formed on the model nanodendrite based on the calculation using FDTD.

for the examination of EM characteristics of diverse metallic nanostructures, such as study of EM property for a two-dimensional nanodendrite structure.[11] They concluded that the dendrite-like structure was able to enhance EM field largely, thereby advantageous for SERS-based measurement.

With the use of FDTD, the calculation of EM field enhancement in a three dimensional (3D) nanodendrite under irradiation of 785 nm laser was performed in this study. It is necessary to note that FDTD is limited for the calculation of EM enhancement for a single-component structure. For this purpose, a 3D Au nanodendrite composed of 8 terraces (branches) arranged every 120° around the column was used as a model structure. Fig. 5 presents the distribution of near-field electrical (a) and magnetic (b) field formed on the model nanodendrite based on the calculation using FDTD. The scale in the electrical field is approximately 5 times larger than that in the magnetic field. The enhancement of electric field mainly occurs on the branches. The traversing laser radiation induces electrical field with varying magnitude mainly on the top and trunk of the branches. In contrast, in the case of magnetic field, the enhancement widely occurs along the column as well as inside the angle between the column and branches. When both electric and magnetic field enhancements are combined, hot spots able to elevate SERS intensity broadly distributed over a whole 3D nanodendrite, thereby suggesting it as a versatile substrate for SERS-based analysis.

SERS measurement using Au–Ag bimetallic nanodendrite

The magnitudes of SERS signals obtained using the bimetallic nanodendrites with different Au–Ag compositions were examined and also compared with that acquired using Au nanodendrite. For this purpose, nanodendrites were allowed in 100 nM 2-NpSH solution for 60 minutes and Raman spectra of the 2-NpSH-adsorbed nanodendrites were measured. Fig. 6 shows baseline-corrected SERS spectra of 2-NpSH collected using eight different Au–Ag bimetallic nanodendrites as well as the Au nanodendrite. The inset shows the SERS intensities at 1378 cm^{-1} (the strongest peak) acquired using each substrate. The top and bottom x-axes correspond to the atomic Ag percentage in the nanodendrites and the molar percentage of Ag^+ in the electrodeposition solution, respectively. Error bars indicate the standard deviation of 10 separate measurements. SERS intensity increased as the Ag concentration

 This journal is © The Royal Society of Chemistry 2014

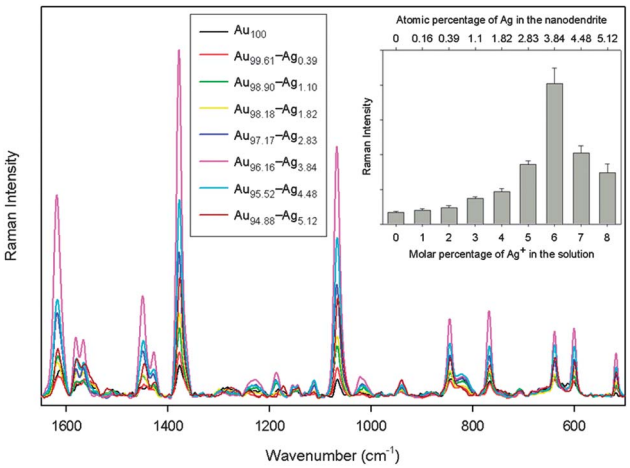

Fig. 6 Baseline corrected SERS spectra of 2-NpSH collected using eight different Au–Ag bimetallic nanodendrites as well as Au nanodendrite. The inset shows the SERS intensities at 1378 cm^{-1} acquired using each substrate. The top and bottom *x*-axes correspond to the atomic Ag percentage in the nanodendrites and the molar percentage of Ag^+ in the solutions used for electrodeposition, respectively.

increased and reached a maximum when atomic Ag concentration was 3.84%. Then, for Ag concentrations higher than 3.84%, it decreased substantially, corresponding to formation of a less characteristic dendrite structure as shown in Fig. 1(d). Increasing the concentration of Ag in the nanodendrite enhanced SERS intensity as long as clear nanodendrite structures maintained; however, the imperfect nanodendrite structures even with the higher Ag concentrations (>3.84%) resulted in a considerable decrease in SERS intensity due to a decrease in the population of hot spots. The enhancement factor calculated for Au nanodendrite and $Au_{96.16}$–$Ag_{3.84}$ nanodendrite were 6.1×10^5 and 6.9×10^6, respectively, so 11-fold strong Raman intensity for $Au_{96.16}$–$Ag_{3.84}$ nanodendrite compared to Au nanodendrite.

Conclusions

We demonstrated that co-electrodeposition of Au and Ag is a simple and effective method to synthesize Au–Ag bimetallic nanodendrite structures. It provided the increased SERS intensity for 2-NpSH in comparison with that composed of only Au. The proposed substrate could be further useful for sensitive analytical measurements based on either SERS or electrochemistry *via* incorporation of a novel sensing scheme on the structure. We are under developing a scheme for fabrication of tri-metallic nanostructures, which are also highly potential in analytical measurements and improving catalytic activity.

Acknowledgements

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (NRF-2012K2A1A2033022).

Notes and references

1 W. Zhang, J. Yang and X. Lu, *ACS Nano*, 2012, **6**, 7397–7405.
2 Y. Yamauchi, A. Tonegawa, M. Komatsu, H. Wang, L. Wang, Y. Nemoto, N. Suzuki and K. Kuroda, *J. Am. Chem. Soc.*, 2012, **134**, 5100–5109.
3 N. L. Netzer, C. Qiu, Y. Zhang, C. Lin, L. Zhang, H. Fong and C. Jiang, *Chem. Commun.*, 2011, **47**, 9606–9608.
4 P. R. Evans, W. R. Hendren, R. Atkinson and R. J. Pollard, *Nanotechnology*, 2008, **19**, 465708.
5 X. Hong, D. Wang, S. Cai, H. Rong and Y. Li, *J. Am. Chem. Soc.*, 2012, **134**, 18165–18168.
6 S. D. Choudhury, R. Badugu, K. Ray and J. R. Lakowicz, *J. Phys. Chem. C*, 2012, **116**, 5042–5048.
7 W. Xia, J. Sha, Y. Fang, R. Lu, Y. Luo and Y. Wang, *Langmuir*, 2012, **28**, 5444–5449.
8 C. Wang, J. Fang, Y. Jin and M. Cheng, *Appl. Surf. Sci.*, 2011, **258**, 1144–1148.
9 L. Zhang, L. Chen, H. Liu, Y. Hou, A. Hirata, T. Fujita and M. Chen, *J. Phys. Chem. C*, 2011, **115**, 19583–19587.
10 S. Besner and M. Meunier, *J. Phys. Chem. C*, 2010, **114**, 10403–10409.
11 T. Qiu, Y. Zhou, J. Li, W. Zhang, X. Lang, T. Cui and P. K. Chu, *J. Phys. D: Appl. Phys.*, 2009, **42**, 175403.
12 X. Chen, C. H. Cui, Z. Guo, J. H. Liu, X. J. Huang and S. H. Yu, *Small*, 2011, **7**, 858–863.
13 K. Jasuja and V. Berry, *ACS Nano*, 2009, **3**, 2358–2366.
14 L. He, T. Rodda, C. L. Haynes, T. Deschaines, T. Strother, F. Diez-Gonzalez and T. P. Labuza, *Anal. Chem.*, 2011, **83**, 1510–1513.
15 J. J. Feng, A. Q. Li, Z. Lei and A. J. Wang, *ACS Appl. Mater. Interfaces*, 2012, **4**, 2570–2576.
16 T. N. Huan, T. Ganesha, K. S. Kim, S. Kim, S. H. Han and H. Chung, *Biosens. Bioelectron.*, 2011, **27**, 183–186.
17 Y. Xia, J. Liu, W. Huang and Z. Li, *Electrochim. Acta*, 2012, **70**, 304–312.
18 S. Cherevko, X. Xing and C. H. Chung, *Electrochem. Commun.*, 2010, **12**, 467–470.

This journal is © The Royal Society of Chemistry 2014